

A Wide-band SWaP System for VSWR and Power Measurement with Temperature Compensation Technique

Nirbhay Kumar Singh
Product development and innovation center
Bharat Electronics Limited
Bengaluru, India
nirbhaykumar@bel.co.in

T. Venkatamuni
Product development and innovation center
Bharat Electronics Limited
Bengaluru, India
tvenkatamuni@bel.co.in

Abstract—Modern high-power systems like Radars and communication need very accurate and temperature invariant systems to measure the transmitted power and VSWR to insure the proper operation of these equipment. This paper discusses the design of a wide-band SWaP (size, weight, and power) system for VSWR and power measurement with very high accuracy and dynamic range with different duty cycles of operation. Measurement accuracy is achieved using digital RF signal processing and voltage mapping techniques.

This paper also discusses the temperature compensation technique for a wide band temperature of operation (-10°C to $+70^{\circ}\text{C}$) with less than or equal to ± 0.3 dB of measurement of accuracy. Finally, the results of L & S - Bands with various power levels and temperatures are presented.

Keywords—SWaP, VSWR, MTBF, Temperature Compensation, ADC, Detectors, Power measurement, Digital RF signal processing, 1G Ethernet.

I. INTRODUCTION

VSWR and Power measurements are the performance parameters for any high-power amplifier system. Measuring or monitoring transmitted and reflected power from phase array antenna ensures that a Radar/communication system operates within its stated power and VSWR limits. Accurate measurement of the received power from phase array antennas in radar systems can also help to identify the distance of friend or foe (IFF, identification of friend or foe) or calibrate receiver sensitivity. In recent times, the various ways of measuring transmitted and reflected power in systems have improved due to integrated circuits (ICs) with power-measurement capability.

Generation of high power could be expensive as frequency increases and also costly to maintain equally for a longer period of time. A fraction of power lost in a radar/communications system, may be due to loss through an active or passive component—can affect the reception of the microwave or radio frequency signal. Most of the military systems have some way of power measurement to observe issues before they occur and forestall operational of systems interruptions.

Most of the all recent new technology developments, whether airborne systems, shipboard, hand-carried or man-carried, have a common requirement: make them smaller, light weight, less power consumption and have better mean

time between failure (MTBF). In the point of view a SWaP system has been designed with very accurate VSWR and power measurement (VPM) through log detector and digital RF signal processing[4].

The VPM consists of three channels RF front end, digital and power supply circuitry as shown in fig 1. The RF front end converts L&S Band signals to Video signals after signal conditioning. High accuracy and wideband log detectors [1] are used in each front end for RF to video conversion. High-resolution ADCs are used to digitize video signals for accurate VSWR and power measurement. Reference, forward & reverse power measurement and VSWR being calculated digitally in FPGA.

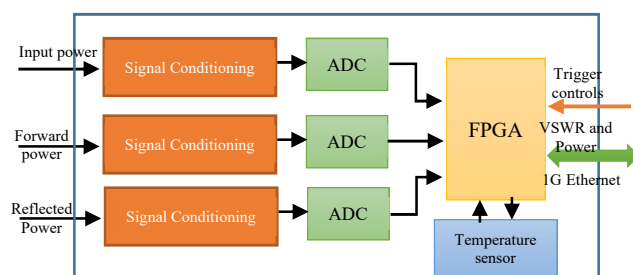


Fig. 1. VPM block diagram

After receiving triggers from the Radar controller (RC), VSWR and POWER measured data (Reference, forward & reverse power) are transferred to RC through 1G Ethernet (UDP) communication. In addition to Ethernet, RS422 interface will be provided for data communication. A temperature sensor is used for the temperature composition of detector response.

II. CONCEPTUAL DESIGN

A. Change in detector response slope and Voltage mapping technique

The detector is used for the RF to video conversion. Detector responses will play a critical role in measuring accuracy.

The equation for the output voltage of detector can be written as [2]

$$V_{\text{OUT}} = m \times (P_{\text{IN}} - P_0) \quad (1)$$

where m is the slope expressed in volts per decibel. P_0 is an intercept point at which the output voltage is 0V.

Calculation of the slope and P_0 are given by using the following equations:

$$m = \frac{V_{OUT(MEASURED)1} - V_{OUT(MEASURED)2}}{P_{IN1} - P_{IN2}} \quad (2)$$

$$P_0 = P_{IN1} - \frac{V_{OUT(MEASURED)}}{m} \quad (3)$$

Once the slope and P_0 are calculated, $V_{OUT(IDEAL)}$ can be calculated using the equation 1, and the error is calculated using the following equation:

$$Error = \frac{V_{OUT(MEASURED)} - V_{OUT(IDEAL)}}{m} \quad (4)$$

For the selected detector, Logarithmic Slope 21 mV/dB and Logarithmic P_0 -88 dBm.

1dB change in the input power, detector output changes 21mV/dB which is very low for the ADC to detect in the noisy condition. To increase the detector slope, operation amplifier (Op-amp) is used as shown in figure 2.

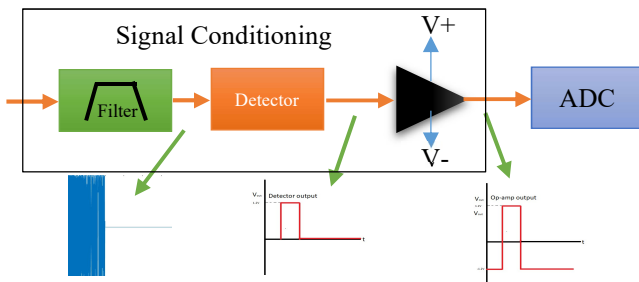


Fig. 2. Signal conditioning with different stages waveforms

Op-amp is used to increase the slope by 2 times as well as shift the waveform in negative voltage for the proper ADC range as shown in figure 3.

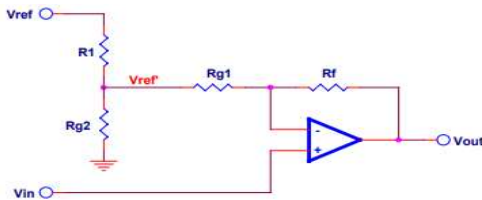


Fig. 3. Non-Inverting op-amp configurations with DC shift

Op-amp is used for voltage amplification by 2 factor and offset by -1.2V

$$V_{out} = V_{in} * \left(\frac{1+R_f}{R_{g1}+R_{g2}} \right) - V_{ref} * \left(\frac{R_f}{R_{g1}+R_{g2}} \right) \quad (5)$$

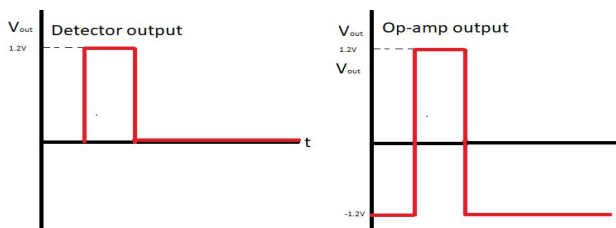


Fig. 4. Detector and Op-amp output with -ve offset of 1.2V

Therefore, using op-amp increases the detector slope to 40mV/dBm which increases the measurement accuracy.

B. Temperature compensation [3] technique for accurate power measurement

Detector output response over temperature with 80dB dynamic range is shown in figure 5. Due to a change in temperature, the output response of the detector changes to 2dB which is very high [5].

To overcome the above problem, the temperature compensation technique is used. Digital signal processing with temperature sensor and lookup in FPGA helps to achieve measurement accuracy.

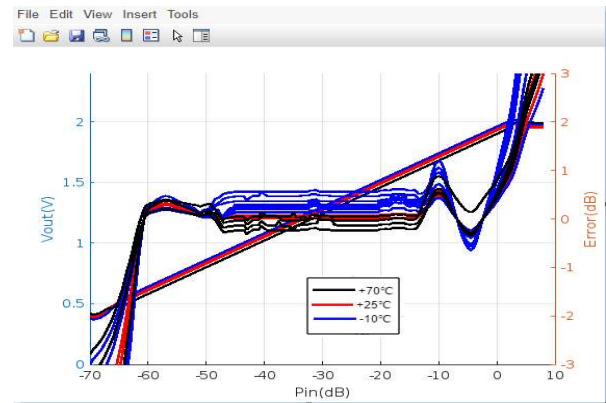


Fig. 5. Detector response over different temperatures

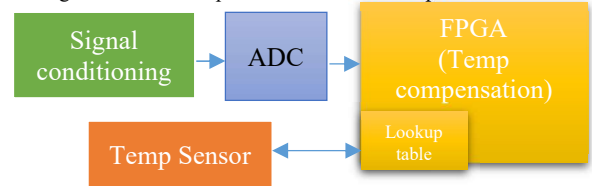


Fig. 6. Temperature compensation technique block diagram

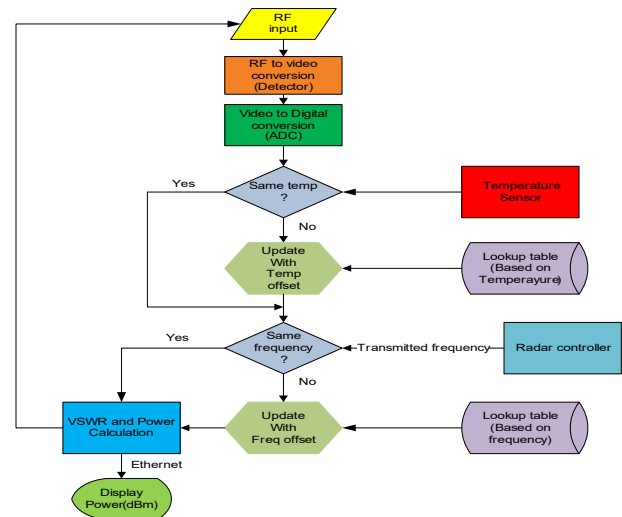


Fig. 7. Temperature and frequency compensation flowchart

Video digital signal has been captured by FPGA and using a temperature sensor and lookup table stored inside FPGA, effect of temperature being compensated as shown to figure 7. After the compensation of response (capture inside FPGA) of detector is shown in figure 8.

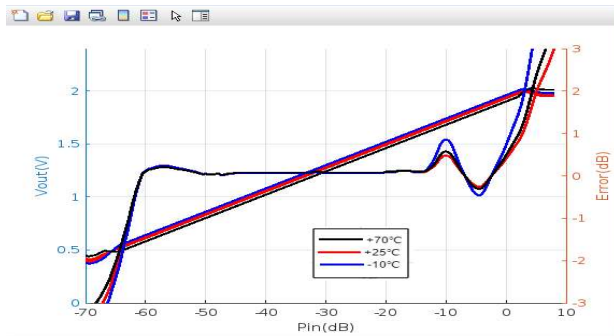


Fig. 8. Detector response after temperature compensation

C. Digital RF signal processing for VSWR and power measurement

Power is calculated based on detector voltage – power response. RF signal is converted to digital using ADC and power has been calculated in FPGA using a lookup table. For VSWR calculation, following equations are used.

$$\text{Reflection coefficient } \Gamma = \frac{\text{Reflected Power(watts)}}{\text{forward power(watts)}} \quad (6)$$

$$\text{Voltage standing wave ration (VSWR)} = \frac{1+|\Gamma|}{1-|\Gamma|} \quad (7)$$

$$\text{Power (dBm)} = 10 * \log\left(\frac{\text{power(watts)}}{1000}\right) \quad (8)$$

III. IMPEMENTATION

VPM module has been developed for the measurement of Power and VSWR for the frequency of operation from 10MHz to 4GHz. It is very lightweight (300 Grams), small in size (200mm*100mm*45mm), and has low power conception. It is a SWaP system for radar and communication applications. High-speed ADCs are used for analog to digital conversion with good dynamic range. ADCs output are digital video signal and capture by parallel processing-based FPGA. Pulse integration has been done in the digital domain to get better accuracy. VHDL-based code has been written for the temperature and frequency compensation as shown in fig 7. Power is calculated in dBm based on the voltage to power plot as shown in fig 8.



Fig. 9. Developed VPM module

Finally, based on the forward power and reflected power from the antenna, VSWR is calculated using equations 6 and 7. Calculated VSWR and power are sent over the ethernet to the radar controller for display. MATLAB based graphical user interface has been created to plot the linearity and error measurement.

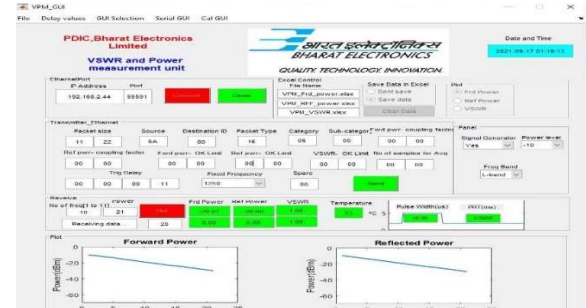


Fig. 10. MATLAB graphical user interface (GUI) for data capture

IV. RESULTS

VPM module is tested for the various band of frequencies and after temperature and frequency compensation, accuracy in power measurement is ± 0.3 dB (max). Plot of detector output for linear frequency modulation (LFM) with frequency 2800MHz and pulse width 500ns is capture through CRO is show in fig 11.

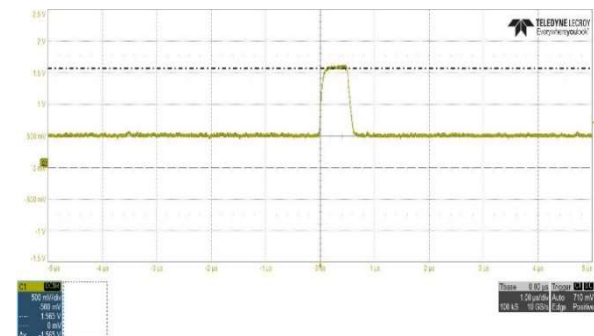


Fig. 11. Detector output with 500ns LFM pulse width

After scaling of detector response using op-amp and with DC offset to get better resolution per dBm power, op-amp output is show in fig 12.

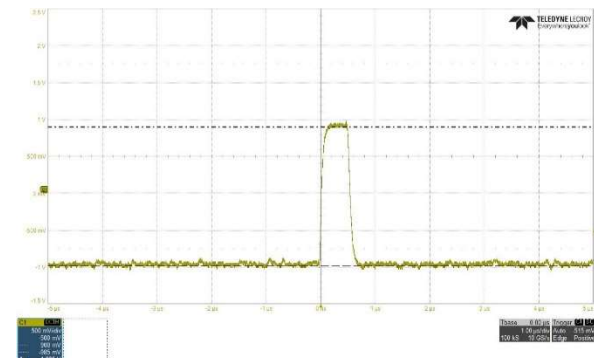


Fig. 12. Op-amp output with 500ns LFM pulse width

Flatness of the signal conditioning response is tested before the detect and show in fig 13. There is 1.5dB variation is observe and compensated using lookup table.

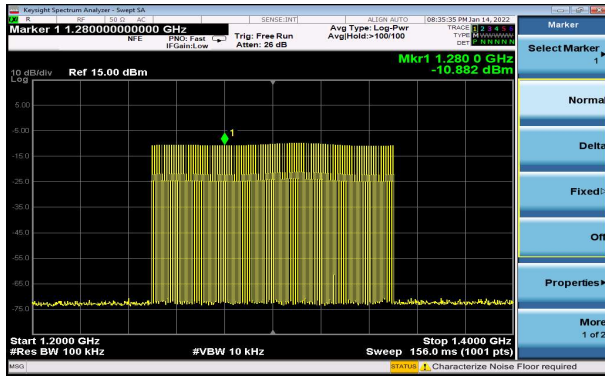


Fig. 13. Detector flatness response for multiple spot of frequencies

VSWR and power are measured inside FPGA and sent through 1G ethernet and captured using MATLAB GUI. Captured data is stored in excel using MATLAB and forward power measurement for the frequent 1250 MHz to 1350 MHz

TABLE II. FORWARD, REFLECTED POWER AND VSWR MEASUREMENT READING

Forward Power		Reflected Power		VSWR	
Input Power	Measured Power	Input Power	Measured Power	Calculated VSWR	Measured VSWR
-10	-10.15	-20	-19.96	1.92	1.98
-11	-10.98	-21	-20.73	1.92	1.96
-12	-11.96	-22	-21.75	1.92	1.95
-13	-13.01	-23	-22.82	1.92	1.95
-14	-14.03	-24	-23.84	1.92	1.95
-15	-14.84	-25	-24.84	1.92	1.92
-16	-15.95	-26	-25.86	1.92	1.93
-17	-16.93	-27	-26.81	1.92	1.94
-18	-18.02	-28	-27.9	1.92	1.94
-19	-19.08	-29	-28.9	1.92	1.94
-20	-20.07	-30	-29.76	1.92	1.98
-21	-21.09	-31	-30.89	1.92	1.99
-22	-22.06	-32	-31.81	1.92	1.99
-23	-23.2	-33	-32.71	1.92	2
-24	-24.2	-34	-33.7	1.92	2
-25	-25.17	-35	-34.73	1.92	1.99
-26	-26.15	-36	-35.7	1.92	2
-27	-26.92	-37	-36.75	1.92	1.96
-28	-28.04	-38	-37.73	1.92	1.97
-29	-29.08	-39	-38.68	1.92	1.98
-30	-30.05	-40	-39.6	1.92	1.99

VPM module is tested with temperature range from -10°C to +55°C and result is shown in table III.

TABLE III. THERMAL CYCLING READING (-10°C TO +55°C)

VPM Temperature Test on 06.09.2021				
Test start time				
Time (PM)	Temperature(°C)	Output	Ref Power	VSWR
2.45	27	66.6	56.2	1.21
2.55	54.3	66.6	56.1	1.21
2.56	54.9	66.8	56.1	1.21
3.01	56	66.7	56.1	1.21
3.05	55.8	66.8	56.1	1.21
3.13	55.3	66.8	56.1	1.2
3.25	55.1	66.9	56.1	1.21
3.36	55	66.8	56.2	1.21
3.55	55	66.8	56.1	1.21
4.15	55	66.8	56.1	1.2
4.45	55	66.8	56.2	1.21
5.18	55	66.8	56.1	1.19
5.38	5	66.5	56.3	1.21
5.45	-7	66.4	56.3	1.21
6.01	-10.7	66.4	56.2	1.21
6.15	-10.9	66.6	56.2	1.21
6.31	-10	66.5	56.4	1.21
6.47	-10	66.4	56.3	1.21
6.63	-10	66.5	56.4	1.22

Module is tested for various test condition and found the very accurate measurement in the power and VSWR.

with 10MHz step are shown in table I. It is observed that the accuracy of measurement is ± 0.3 dB (max).

TABLE I. FORWARD POWER MEASUREMENT READING

Forward Power										
Input Power	Input frequency									
	1250 MHz	1260 MHz	1270 MHz	1280 MHz	1290 MHz	1300 MHz	1310 MHz	1320 MHz	1330 MHz	1340 MHz
-10	-9.93	-10.07	-10.19	-10.21	-10.10	-10.29	-9.83	-9.61	-9.86	-9.89
-11	-10.91	-11.12	-10.94	-10.98	-11.02	-11.02	-11.00	-10.92	-11.01	-10.98
-12	-11.92	-12.14	-11.94	-11.94	-12.00	-11.99	-11.96	-11.90	-12.17	-11.94
-13	-13.03	-13.25	-12.98	-13.05	-13.10	-13.10	-13.03	-12.96	-13.05	-13.05
-14	-14.03	-14.28	-13.99	-13.97	-14.12	-13.87	-14.03	-13.95	-14.26	-14.23
-15	-14.99	-15.24	-14.95	-14.95	-15.16	-15.06	-15.03	-14.95	-15.02	-14.97
-16	-15.95	-16.22	-15.90	-15.97	-16.12	-16.08	-16.03	-15.95	-15.99	-15.95
-17	-16.93	-17.19	-16.89	-16.93	-17.13	-17.10	-17.04	-17.10	-17.22	-17.02
-18	-18.04	-18.28	-18.00	-18.11	-18.21	-18.20	-18.15	-18.12	-18.24	-18.06
-19	-19.02	-19.26	-19.04	-19.11	-19.26	-19.20	-19.29	-19.11	-19.15	-19.02
-20	-20.00	-20.29	-20.04	-20.12	-20.28	-20.20	-20.17	-20.37	-20.16	-20.07
-21	-20.98	-21.25	-21.15	-21.10	-21.14	-21.22	-21.16	-21.11	-21.18	-21.05
-22	-21.85	-22.27	-22.46	-22.10	-22.23	-22.20	-22.11	-22.05	-22.14	-22.05
-23	-23.09	-23.18	-23.16	-23.19	-22.92	-23.17	-23.22	-23.12	-23.19	-23.16
-24	-24.12	-24.15	-24.18	-24.21	-24.09	-24.07	-23.96	-23.89	-24.17	-24.14
-25	-25.08	-25.29	-25.12	-24.94	-25.06	-24.85	-24.95	-24.80	-24.92	-25.13
-26	-26.10	-26.19	-26.12	-25.93	-26.41	-26.01	-25.92	-25.85	-25.89	-26.08
-27	-27.09	-27.21	-26.88	-26.94	-27.04	-27.00	-26.99	-26.82	-26.86	-26.82
-28	-28.12	-28.50	-28.02	-27.98	-28.07	-28.12	-28.04	-27.92	-27.94	-27.85
-29	-29.18	-29.29	-29.04	-29.02	-29.09	-29.07	-29.07	-28.92	-28.91	-28.86
-30	-29.93	-30.22	-30.05	-30.17	-30.04	-30.08	-30.04	-30.04	-29.90	-29.85

Forward, reflected and VSWR for different input power are show in the table II.

V. CONCLUSION

Log detectors can be easily used to measure gain and return loss because of their linear-in-dB transfer function. But at the same time, they are sensitive to temperature changes. In-room temperature, accuracy can be achieved without calibration or compensation for their response. By using change in detector response slope & Voltage mapping and temperature compensation technique, very high measurement precision is achievable. In all cases, careful input power level planning is necessary so that the log detectors are driven at input power levels that offer good linearity in the output voltage and temperature stability.

This paper discussed the VSWR and Power measurement using RF-Digital signal processing. The highly accurate ADC and FPGA-based module is designed and tested over extreme operating environmental conditions. By using the Frequency compensation technique, wideband of operation frequency has been achieved. Finally, the implementation of the temperature compensation technique temperature stability is presented and the results are analyzed.

ACKNOWLEDGMENT

We are thankful to Mr. G Rambabu (Sr-DGM RFMW-PDIC) for sharing his knowledge with us during the development of this new technology project. We would also like to show our gratitude to Mr. Suresh Kumar K V, General manager-PDIC for providing continuous support and opportunity to work on this new technology project.

REFERENCES

- [1] Eamon Nash Applications Engineer, Analog Devices, Inc. "Measurement and Control of RF Power (Part I)".
- [2] Eamon Nash Applications Engineer, Analog Devices, Inc. "Measurement and Control of RF Power (Part II)".
- [3] Eamon Nash Applications Engineer, Analog Devices, Inc. "Measurement and Control of RF Power (Part III)".
- [4] EAMONNASH, Analog Devices, Wilmington, MA, "Measuring VSWR and Gain in Wireless Systems".
- [5] Liu Yuanshang, Xu Dawang, Li Jinshan, Li Qiang, "Automatic Temperature Compensation Technology of Peak Power Sensors" IEEE, ICMMT 2020

AUTHORS BIODATA



Nirbhay Kumar Singh- Obtained his B.Tech degree in Electronics and Communication Engineering from Bundelkhand Institute of Engineering and Technology, Jhansi. He joined BEL in 2015 and currently working as Senior Engineer at

Product Development and Innovation Centre (BEL), Bengaluru. He has worked on waveform generation for high resolution Radar, Direct RF Sampling receivers, Adaptive power amplifier linearizer, Wide band Digital transceiver, IFF receiver, Control of Instrument using labview.



T.Venkatamuni – Obtained his B.Tech from Sri Venkateswara University College of Engineering, Tirupati in 1999. Also, he has obtained M.E in Communication Systems from Regional Engineering College

Tiruchirappalli in 2001. He joined Central Research Laboratory-Bharat Electronics Ltd. Bangalore, in 2001 and since then he is working in the RF and Microwave group as a Member Research Staff. Presently he is working as a Deputy General Manager (DGM) at Product Development and Innovation Centre (BEL), Bengaluru. He has worked on Adaptive signal processing for anti-jamming, Power amplifiers linerarizers and FMCW radars reflected power cancellation. His area of interests includes handheld RF Transceivers and ultra-low phase noise synthesizers.